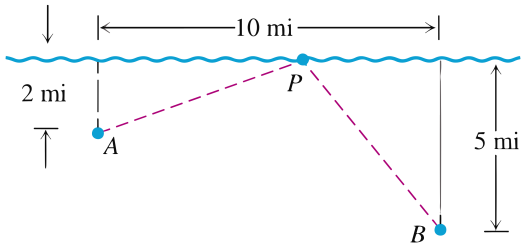
Calculus I

Tutorial #5 (28 Oct – 01 Nov 2024)

Topics:

- Extreme values of functions
- The Mean Value Theorem
- The first derivative test
- Concavity and curve sketching

Problem 1. *(Locating a pumping station)* Two towns lie on the south side of a river. A pumping station is to be located to serve the two towns. A pipeline will be constructed from the pumping station to each of the towns along the line connecting the town and the pumping station. Locate the pumping station to minimize the amount of pipeline that must be constructed.



Problem 2. Does the function $f(x) = \sqrt{x(1-x)}$ satisfy the hypotheses of the Mean Value Theorem on $[0, 1]$? Justify your answer.

In Problems 3–4, plot the zeros of each polynomial on a line together with the zeros of its first derivative.

Problem 3.

(a) $y = x^2 - 4$

(b) $y = x^2 + 8x + 15$

Problem 4.

(a) $y = x^3 - 3x^2 + 4 = (x + 1)(x - 2)^2$

(b) $y = x^3 - 33x^2 + 216x = x(x-9)(x-24)$.

Problem 5. Use Rolle's Theorem to prove that between every two zeros of $x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$ there is a zero of $nx^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + a_1$.

Problem 6. Show that the functions have exactly one zero in the given interval.

a) $f(x) = x^4 + 3x + 1, \quad x \in [-2, -1]$.

b) $r(\theta) = \theta + \sin^2\left(\frac{\theta}{3}\right) - 8, \quad \theta \in \mathbb{R}$.

Problem 7. *(Temperature change)* It took 14 sec for a mercury thermometer to rise from $-19°C$ to $100°C$ when it was taken from a freezer and placed in boiling water. Show that somewhere along the way the mercury was rising at the rate of $8.5°C/sec$.

Problem 8. For the functions

$$f(t) = 12t - t^3, \quad -3 \le t < \infty \quad \text{and} \quad h(x) = \frac{x^3}{3} - 2x^2 + 4x, \quad 0 \le x < \infty$$

(a) Identify the function's local extreme values in the given domain, and say where they are assumed.

(b) Which of the extreme values, if any, are absolute?

(c) Support your findings with a graphing calculator or computer grapher.

Problem 9. If b, c, and d are constants, for what value of b will the curve $y = x^3 + bx^2 + cx + d$ have a point of inflection at $x = 1$? Justify your answer.

Problem 10. *(Quadratic curves)* What can you say about the inflection points of a quadratic curve $y = ax^2 + bx + c, a \neq 0$? Justify your answer.

THE END

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Answer to selected problems

1. $\left(\frac{20}{7}, 0\right)$
9. -3
10. Doesn't exist.